

E2EAI: End-to-End Deep Learning Framework for Active Investing

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ABSTRACT

Active investing aims to construct a portfolio of assets that are expected to be relatively profitable in the markets. A popular strategy involves the use of factor-based methods. Recently, efforts have increased to apply deep learning to identify “deep factors” that could provide more active returns or promising pipelines for asset trend prediction. However, the question of constructing an active investment portfolio via an end-to-end deep learning framework (E2E) remains largely unexplored in existing research. In this paper, we are the first to propose an E2E approach that encompasses nearly the entire process of factor investing, including factor selection, combination, stock selection, and portfolio construction. A key challenge we address is the potential divergence in the directions of deep factors across different horizon lengths, which can create conflicts in the learning process of our multi-task learning model, E2EAI. To overcome this, we design a directional recovery algorithm that ensures consistent learning across tasks. Extensive experiments on real stock market data demonstrate the effectiveness of our end-to-end deep learning framework in active investing. Our approach not only enhances the potential returns of active investment strategies but also provides a comprehensive solution for managing multi-task learning conflicts in the context of deep learning-based factor investing.

KEYWORDS

deep learning, quantitative investment, financial data.

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1 INTRODUCTION

Active investing, aimed at generating returns exceeding a market benchmark, is a dominant investment style in financial markets. Factor investing, a common quantitative approach, leverages factors as predictive tools for future returns. This process involves manually creating factor databases, selecting useful factors through testing,

and building a multifactor model for stock selection. An active portfolio, seeking to outperform a major market index, is then constructed. Despite this sequential process, these steps are often optimized in isolation, leaving end-to-end active investment (E2E) largely unexplored in existing works [11, 28, 29].

Current practices in active investing often result in sub-optimal portfolios for several reasons. First, factor selection rules of thumb, such as requiring a certain absolute value for the information coefficient (IC) and its z-score (ICIR), may neglect factors that, while useful for identifying attractive stocks, don’t exhibit a linear correlation with expected returns. Second, standard factor selection does not ensure factor diversity, potentially resulting in a portfolio overly exposed to a single risk factor. Third, portfolio construction using a multifactor model and maximizing weighted factor values doesn’t directly lead to optimal portfolios. Stocks selected based on their scores may not yield the most profitable portfolio, as often, less attractive stocks generate higher returns. The goal should be to maximize portfolio return subject to certain constraints on stock weighting. Lastly, the use of isolated modules, each with a different learning objective, may detract from achieving a cohesive investment objective. A more systematic approach would involve an end-to-end (E2E) framework with a unified learning objective, ensuring all modules work towards the same goal.

To overcome the limitations of sequentially using isolated modules in existing deep neural network applications, we introduce a novel End-to-End Active Investing framework (E2EAI). This framework unifies all investment process modules—factor selection, stock selection, and portfolio construction—under a single learning objective, such as optimizing portfolio returns under certain weighting or risk constraints. Distinctively, we develop a gated attention block for both factor selection and portfolio construction. Unlike conventional practice that uses IC or ICIR thresholds for factor selection, our gated-attention mechanism identifies the most contributive factor for portfolio construction. Furthermore, we employ a deep multifactor model that learns a deep factor via attention modules based on a multi-relationship stock graph, considering intra-sector and cross-sector influences. Portfolio construction then relies on the exposure of deep factors and the company’s market and fundamental context learned from original inputs. Crucially, our entire learning framework operates under a unified objective, setting it apart from existing methodologies.

However, one of the key challenges in end-to-end active investing lies in the potential directional conflicts of deep factors. This is a multifaceted problem, as active investing involves multiple learning objectives across various horizons. Deep factors associated with each of these horizons may have different directions, signifying that the correlation between each deep factor and their future returns

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under each horizon would vary. For instance, a certain deep factor under a particular horizon might have a positive correlation with the returns over the same horizon, while under other horizons, the correlation might be negative. This discrepancy can lead to learning conflicts during the end-to-end learning process, complicating the construction of effective active investment strategies.

In response to this challenge, we introduce a novel Directional Recovery Algorithm. Central to this algorithm is a direction buffer, a unique mechanism designed to track and retain the long-term relationship between each deep factor and its future return across different horizons. By preserving a consistent understanding of these relationships, the direction buffer effectively mitigates the conflicts arising from divergent factor directions. This innovative approach facilitates the creation of more robust and efficient end-to-end active investing strategies, representing a significant advancement in the field.

While an E2E framework can deliver promising results in active investing, portfolio managers often seek financial logic and insights into the workings of deep learning, particularly how original factors influence the deep factor used for portfolio construction. Consequently, an E2E framework must also offer interpretability. To this end, we develop a linear and directional estimator within the deep multifactor module to glean insights and elucidate the logic behind original factors from a linear representation standpoint. Contrary to existing work [16, 17, 26], we introduce a directional attention mechanism capable of identifying the contribution of original factors and indicating their direction as positive or negative contributions. A similar directional buffer can also be applied to the deep factor, providing a direct indication of whether its correlation with future return is positive or negative.

To demonstrate the effectiveness of our approach, we perform a comprehensive study using real data from over 2800 constituent stocks across three broad-based indices in Chinese stock markets. Our E2E framework consistently outperforms existing investment pipelines with separately optimized modules. The deep factor learned by our E2E method also delivers superior portfolios compared to existing multifactor models. In summary, our work makes significant contributions by:

- (1) Introducing the first E2E framework for active investing that uses original data for portfolio construction;
- (2) Proposing a novel loss design incorporating a directional multiplier to determine the direction of the deep factor, recover the original factor contributions, and ascertain the historical cross-sectional factor returns of the deep factor;
- (3) Developing a training algorithm that includes a global optimizer for training the E2E framework and a local optimizer for calculating the historical factor returns of the deep factor with a directional recovery.

2 RELATED WORKS

Deep Multifactor Model Incorporating Stock Relationship. The multifactor model has received considerable attention from researchers and academics for several decades to determine the exact nature of the common factors that influence risk and return in various assets and markets [14]. A large body of work has shown that deep or nonlinear models perform better than linear models in this

context [9, 16, 17]. Moreover, cross-sectional factors can provide a better explanation for average stock returns [6], inspiring us to develop a deep learning architecture to learn cross-sectional factors. With the rapid development of deep learning and its advantages in learning nonlinear relationships from Big Data in finance [8], graph neural networks have been successful in solving a variety of finance problems [24]. These include stock movement prediction [3, 11, 28], event-driven prediction [29], and risk management [10]. In our work, we design a relational neutralization block based on a gated graph attention mechanism to learn the multi-relational stock relationship from different perspectives [22]. However, it is important to note that while some recent deep learning methods have also been applied to stock prediction, using multi-scale [4], multi-patterns [23, 31], regime switching [1, 12, 20, 27], or additional information [30] in the modeling, these are not suitable counterparts for investigating end-to-end active investing. Our target is to alleviate the challenging directional conflicts that occur during multi-task learning in E2EAI. Thus, these deep learning methods may not be covered in the comparison.

Interpretation of deep factors. Although deep learning methods perform well in learning deep factors in terms of the multi-factor model, the lack of transparency and limited interpretability of the deep factor make it difficult for institutional investors to apply a model that operates in a black-box manner in actual investment practice. One remedy is to use layer-wise relevance propagation (LRP) [2], a method that highlights which input features it uses to support its prediction [15], to capture the linear relationship between the input factors and the prediction [13, 17]. Later, [26] provides a more direct way to learn the linear contribution of each input factor to the deep factor via an attention mechanism. However, these methods neglect the directional property of the original factor's contribution to the deep factor, *i.e.*, the contribution of an original factor can be positive or negative. Without considering the different directions of the contribution, the linear approximation of the original factors in [26] may ignore the factor that contributes negatively to the deep factor, resulting in a "biased" approximation. In this paper, we propose a directional linear approximation of the deep factor via an attention module and a directional buffer to account for both the positive and negative contribution of the original factor to the deep factor.

Portfolio construction. In existing work, portfolio construction is usually performed by the ad hoc methods, *e.g.*, dividing stocks into ten groups with equal numbers *w.r.t* their factor exposures and selecting the top or bottom group as the portfolio [5, 26]. In our work, we use a soft allocation strategy to simultaneously select attractive stocks and assign them an appropriate weight in the portfolio. Some applications use deep learning methods to solve the portfolio allocation problem [19, 25] in an end-to-end way. However, they solve the problem only in the context of portfolio construction, which is one and the last module in our E2E framework. To best of our knowledge, we are the first to solve the problem of active investing in a holistic but also E2E way.

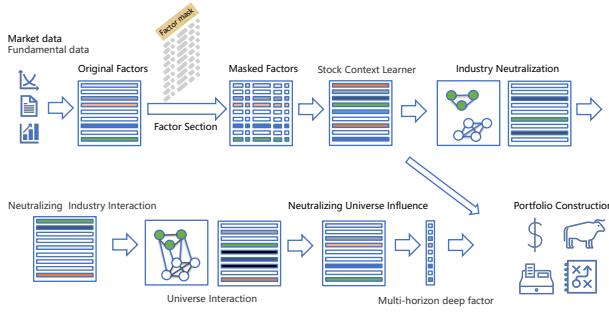


Figure 1: The pipeline of E2E active investing framework.

3 PRELIMINARIES

In the following section, we delve into the preliminaries of our study, setting the groundwork by defining the active investing problem and introducing the concept of a stock graph.

Definition 3.1. The problem of active investing is defined as follows: Given a set of original factors $\mathcal{F}_t = \{f_{it} \in \mathbb{R}^n, i = 1, \dots, m\}$ with m being the number of factors. Each factor is a handcrafted characteristic of n stocks, with values changing over time. These factors can be engineered from market data or fundamental data (which come from financial reports). One should learn a stock allocation $\mathbf{w}_t = [w_{1t}, \dots, w_{nt}]^T$ to maximize the future portfolio return from time t to $t+k$, $r_p^{t+k} = \mathbf{w}_t^T \hat{\mathbf{r}}_{t+k}$, with a set of constraints $\mathcal{C}$. $\hat{\mathbf{r}}_{t+k} = \mathbb{E}[\mathbf{r}_{t+k} | \mathcal{F}_t]$ is the expected return of each stock in the stock pool and $\mathbf{r}_{t+k} \in \mathbb{R}^n$. Some common constraints are: 1) $\sum_{j=1}^n w_{jt} = 1$; 2) $0 \leq w_{jt} \leq u$, where u is the upper bound of a stock allocation with $0 < u \leq 1$.

Definition 3.2. The stock graph [26, 29] is defined as a directed and dynamic graph as, $\mathcal{G}_t = \langle \mathcal{S}_t, \mathcal{R}_t, \mathcal{M}_t \rangle$, where $\mathcal{S}_t$ denotes the set of constituents of a broad-based stock index and $\mathcal{R}_t$ is the set of relations between two stocks at time t . $\mathcal{M}_t$ is the set of adjacent matrices. For an adjacent matrix $\mathbf{M}_t^r \in \mathcal{M}_t$ of relation $r_t \in \mathcal{R}_t$, where $\mathbf{M}_t^r \in \mathbb{R}_+^{|\mathcal{S}_t| \times |\mathcal{S}_t|}$, $\mathbf{M}_t^r(i, j) = 1$ means that there is a relation r from stock s_j^t to stock s_i^t and $\mathbf{M}_t^r(i, j) = 0$ means that there is no such relation.

4 METHOD

In this work, we develop an E2E deep learning framework to find the cross-sectional factors that can consistently explain the average stock returns well, making good use of the relationships between stocks and the insights from finance. (see Fig. 1).

4.1 Factor Selection

The classical way in factor selection is to select factors that meet the IC or ICIR threshold using the rule of thumb. However, this can have two drawbacks: 1) It only considers the isolated and linear correlation between a single factor and the realized return. 2) It is a static selection procedure where the candidate list remains unchanged over a period of time. To mitigate the drawbacks of the existing selection practice, we develop a non-linear and dynamic factor selection module using an attention mechanism. The factor

selection module is defined as the follows

$$\mathbf{M}_f^t = \mathbb{1}_{\geq \gamma_f} (\text{softmax}(\text{MLP}(\mathbf{F}_o^t))), \quad (1)$$

$$\mathbf{F}^t = \mathbf{M}_f^t \mathbf{F}_o^t, \quad (2)$$

where $\mathbb{1}_{\geq \gamma_f}(\cdot)$ denotes an attention gate that forces the focus of the selection module to higher attention factors and γ_f is the attention lower bound for an original factor. $\mathbf{F}_o^t \in \mathbb{R}^{n \times m}$ is a matrix representing m factors w.r.t n stocks.

4.2 Deep Multifactor Model

4.2.1 Stock Context Encoder. We define n as the number of constituents inside a broad-based stock index, and m as the number of types of stock context that covers fundamental, trading information and analysts' estimates.

$$\mathbf{C}_t = \text{MLP}(\text{BatchNorm}(\mathbf{F}_t)), \quad (3)$$

where $\mathbf{F}_t \in \mathbb{R}^{n \times m}$ is a matrix represents $m_s (m_s \leq m)$ factors selected from original factors w.r.t n stocks, and $\mathbf{C}_t \in \mathbb{R}^{n \times m_1}$ is the stock context matrix with m_1 hidden features extracted from the original factors. Batch normalization [7] is a deep learning counterpart to *z-score normalization*, which is generally applied in data preprocessing for factor engineering [26].

4.2.2 Relational Neutralization Block. As a rule of thumb in investment practice, factor processing usually applies industry neutralization and size neutralization to remove the effect of industry and capitalization. Therefore, we design a relational neutralization block to remove the effect of different types of relationships. To encode the stock relationships, we apply Graph Attention Network (GAT) [22] to predefined stock graphs and its corresponding stock context, which is a common paradigm in existing works [24, 26, 28]. The relational neutralization block (RNB) is defined as

$$\tilde{\mathbf{C}}^t = \text{RNB}(\mathbf{C}_t; \mathcal{G}_t) = \mathbf{C}_t - \text{GAT}(\mathbf{C}_t, \mathbf{M}_t^r), \quad (4)$$

where $\mathbf{C}_t \in \mathbb{R}^{n \times m_1}$ is the stock context, $\mathcal{G}_t$ is the given stock graph at time t and $\mathbf{M}_t^r$ is its corresponding adjacent matrix representing the stock relationship r . Similar to [26], we have two neutralized stock context corresponding to inner-industry and cross-industry relationship, $\tilde{\mathbf{C}}_I^t$ and $\tilde{\mathbf{C}}_U^t$, i.e.

$$\tilde{\mathbf{C}}_I^t = \text{RNB}(\mathbf{C}_t; \mathcal{G}_I^t), \quad (5)$$

$$\tilde{\mathbf{C}}_U^t = \text{RNB}(\tilde{\mathbf{C}}_I^t; \mathcal{G}_U^t), \quad (6)$$

where $\mathcal{G}_I^t$ is the industry graph and $\mathcal{G}_U^t$ is the cross-industry graph.

4.2.3 Learning Deep Factors for Multiple Horizons. We design K output heads for learning ultimate factors corresponding to multiple forward periods be aware of practitioners, i.e. k -forward trading days, where $k = 3, 5, 10, 15, 20$. Formally, we learn the deep factor $\mathbf{f}_k^t \in \mathbb{R}^n$ that explains the future information on the k -forward trading day from different granularity: stock original context $\mathbf{C}^t$, industry neutralized context $\tilde{\mathbf{C}}_I^t$, and universe-neutralized context $\tilde{\mathbf{C}}_U^t$:

$$\mathbf{f}_k^t = \text{LeakyReLU}(\mathbf{W}_{[k]}^T (\mathbf{C}^t || \tilde{\mathbf{C}}_I^t || \tilde{\mathbf{C}}_U^t)), \quad (7)$$

where $\mathbf{W}_{[k]}^T$ is a single-layer feed-forward neural network for k -forward trading days and $||$ represents concatenation.

4.3 Portfolio Construction on Automatic Stock Selection

4.3.1 Directional Buffer. To determine the direction of the contribution of each original factor to the deep factor in the deep multifactor module, we develop a directional buffer to “memorize” the directions of each factor. This direction is defined by the sign of the accumulated linear correlations between the original factor and the deep factor in the training period: $\mathbf{d}_k^{[i]} = (d_1, \dots, d_m)$, with $\mathbf{d}_k^{[i]} = \text{sgn} \left(\sum_{t \in \mathcal{T}} \text{IC}(\mathbf{f}_{s,i}^t, \mathbf{f}_k^t) \right)$, where sgn denotes a sign function, $\mathbf{f}_{s,i}^t \in \mathcal{R}^n$ is the i -th factor in $\mathbf{F}^t$. Since we also have a similar directional buffer for the deep factor and the future return across varying prediction horizons, the directional sign stabilizes after sufficient training steps. Consequently, the original directional buffer can also be expressed as $\mathbf{d}_k^{[i]} = \text{sgn} \left(\sum_{t \in \mathcal{T}} \text{IC}(\mathbf{f}_{s,i}^t, \mathbf{r}_k^t) \right)$, where $\mathbf{r}_k^t \in \mathcal{R}^n$ represents the future return for k day ahead. The details of the direction buffer is shown in Algo. 1.

4.3.2 Interpretation of Deep Factor via Directional Attention Module. We introduce directional factor attention module to investigate the positive or negative contribution of original input factors or features to the deep factor. We employ attention mechanism [21] to attend our deep factor to learn the token importance of the original features. The normalized attention weight through a softmax function illustrates “how much information influx in the deep factor from an original feature”.

$$\mathbf{A}_k^t = \text{softmax} \left(\text{LeakyReLU} \left(\mathbf{W}_{a,k}^T \mathbf{F}_t \right) \right), \quad (8)$$

$$\bar{\mathbf{a}}_k^t = \frac{1}{n} \sum_{i \in n} \mathbf{a}_{ik}^t, \quad (9)$$

where $\mathbf{W}_{a,k}^T \in \mathbb{R}^{n \times n}$ is a single-layer feed-forward neural network corresponding to the k -forward trading day. $\mathbf{A}_k^t \in \mathbb{R}^{n \times m}$ is the attention weight matrix for the deep factor $\mathbf{f}_k^t$ and $\mathbf{a}_{ik}^t$ is i -th column of $\mathbf{A}_k^t$. Here, $\mathbf{a}_k^t$ is its corresponding attention weight vector. We define the linear approximation of deep factor, $\hat{\mathbf{f}}_k^t \in \mathbb{R}^m$, as

$$\hat{\mathbf{f}}_k^t = \mathbf{F}_t^T \left(\bar{\mathbf{a}}_k^t \circ \mathbf{d}_k \right), \quad (10)$$

where $\circ$ denotes Hadamard product, $\mathbf{d}_k \in \mathbb{R}^m$ is the directional buffer of original factors w.r.t horizon k . When $\hat{\mathbf{f}}_k^t$ is very close to $\mathbf{f}_k^t$, the attention weight $\bar{\mathbf{a}}_k^t$ can interpret the portion of quantity comes from the original input factors.

4.3.3 Learn a Deep Portfolio with Automatic Stock Selection. We develop a gated additive attention mechanism for automatic stock selection based on the deep factors generated by the previous module and the market embedding learned from all stock contexts. The attention score for all stocks in the universe is defined by Eq. 11 and the allocation of the selected stocks is shown in Eq. 12:

$$\mathbf{A}_{p,k}^t = \text{softmax} \left(\text{LeakyReLU} \left(\mathbf{W}_{a,k}^T \left(\mathbf{C}_t || (\mathbf{f}_k^t \circ \mathbf{d}_{f,k}) \right) \right) \right), \quad (11)$$

$$\mathbf{W}_k^t = \text{softmax} \left(\mathbb{1}_{\geq \gamma_p} (\mathbf{A}_{p,k}^t) \right), \quad (12)$$

where $\gamma_p \in (0, 1]$ is the attention lower bound, $\mathbf{d}_{f,k}$ is the directional buffer of deep factor. If the attention weight of a stock does not

Algorithm 1 Algorithm for Training E2EAI

Input: Training data $\mathcal{D}^{\text{train}}$, model parameters Φ, Ψ

Output: Φ^*, Ψ^*

```

1: Randomly initialize  $\Phi, \Psi$ 
2: while not meet the stopping criteria do
3:    $\mathcal{L}_{\text{train}} \leftarrow 0, l_f \leftarrow 0, b_f \leftarrow 10^{-6}$ , and  $b_i \leftarrow 10^{-6}, i = 1, \dots, m$ 
4:   while  $\mathcal{D}^{\text{batch}} \leftarrow \mathcal{D}^{\text{train}}$  do
5:     Obtain the deep factors via E2EAI:  $\mathbf{f}^{\text{batch}}, \mathbf{w}^{\text{batch}} \leftarrow \Phi(\mathcal{D}^{\text{batch}})$ 
6:      $l_f \leftarrow 0$ 
7:      $d_f = \text{sgn}(b_f)$ , where  $b_f \leftarrow b_f + \text{sum} \left( \text{IC}(\mathbf{r}^{\text{batch}}, \mathbf{f}^{\text{batch}}) \right)$ 
8:      $d_i = \text{sgn}(b_i)$ , where  $b_i \leftarrow b_i + \text{sum} \left( \text{IC}(\mathbf{r}^{\text{batch}}, \mathbf{f}_{s,i}^{\text{batch}}) \right), i = 1, \dots, m$ 
9:     while  $k \leftarrow 1$  to  $K$  do
10:       $l \leftarrow 0$ 
11:      while  $i \leftarrow 1$  to  $N_{\text{iter}}$  do
12:        Calculate the regression loss  $l_i$  of cross-sectional model for deep factor  $\mathbf{f}_k$ , where  $\psi_k$  is its cross-sectional coefficient.
13:         $l \leftarrow l - l_i \cdot d_f$ 
14:        Update  $\psi_t$  via the cross sectional optimizer  $\gamma_1$  by minimizing  $l$ 
15:      end while
16:       $l_f \leftarrow l_f - \text{mean}(\psi_t \circ d_f)$ 
17:    end while
18:    Calculate the loss for factor stability  $l_s$  and the loss for attention estimate  $l_e$ , where  $l_e$  is calculated based on the directional buffer:  $d_i, i = 1, \dots, m$ .
19:    Then, update the train loss:
20:     $\mathcal{L}_{\text{train}} \leftarrow \mathcal{L}_{\text{train}} + l_p + \lambda_e l_e + \lambda_s l_s + \lambda_f l_f$ 
21:    Update  $\Psi$  via the global optimizer  $\gamma_2$ .
22:  end while
23: end while
```

satisfy this requirement, then it would not be considered in the portfolio.

4.4 Loss Design

Now we ponder the design of loss function to fulfill the learning objectives of our E2E active investing. Specifically, the general loss function is

$$\mathcal{L} = \mathcal{L}_p + \lambda_s \mathcal{L}_s + \lambda_f \mathcal{L}_f + \lambda_e \mathcal{L}_e, \quad (13)$$

where $\mathcal{L}_p, \mathcal{L}_s, \mathcal{L}_f$, and $\mathcal{L}_e$ are the loss terms associated with the portfolio, stability, factor returns, and attention estimates, respectively. λ 's are balancing coefficients. In what follows, we denote $\mathcal{T}$ as the set of trading days in the training period and $\mathcal{K}$ as the set of future horizons.

4.4.1 Loss Related to Portfolio. A fundamental objective in portfolio construction is to maximize expected return without violating the weighting constraints on equity allocation. Based on this objective, we express the loss function of our portfolio in two terms, i.e., loss of portfolio $\mathcal{L}_p$ return $\mathcal{L}_{\text{ret}}$ and the loss of weight $\mathcal{L}_{\text{up}}$ penalizing the allocation exceeding its upper bound θ , e.g., $\theta = 0.10$,

where

$$\mathcal{L}_{ret} = -\frac{1}{|\mathcal{T}||\mathcal{K}|} \sum_{k \in \mathcal{K}} \sum_{t \in \mathcal{T}} \mathbf{w}_k^t \mathbf{r}_k^t, \quad (14)$$

and

$$\mathcal{L}_{up} = \frac{1}{|\mathcal{T}||\mathcal{K}|} \sum_{k \in \mathcal{K}} \sum_{t \in \mathcal{T}} (\mathbf{w}_k^t - \theta). \quad (15)$$

In short, $\mathcal{L}_p = \mathcal{L}_{ret} + \mathcal{L}_{up}$.

4.4.2 Maximizing Factor Stability. To enhance the robustness and reliability of financial models, our work aims to maximize the stability of predictive factors, which is quantified through the Information Coefficient Information Ratio (ICIR). The ICIR, defined as the mean of IC divided by its standard deviation, is a prevalent metric for evaluating factor stability [11, 26]. We use c_k to denote the ICIR of the deep factor $\mathbf{f}_k^t$, where c_k measures the stability of the predictive power of the deep factor $\mathbf{f}_k^t$ for the future return over the next k trading days $\mathbf{r}_{t+k}$. Thus, since ICIR is positively correlated to stability, the stability loss is defined as

$$\mathcal{L}_e = \frac{1}{|\mathcal{K}||\mathcal{T}|} \sum_{k \in \mathcal{K}} \sum_{t \in \mathcal{T}} d_k c_k^t, \quad (16)$$

where d_k is the directional indicator with value of 1 or -1. We enhance the precision of our stability assessment by integrating a directional buffer in our stability loss computation, multiplying each ICIR term by its associated directional indicator, d_k . This ensures the consistent directional tendencies are factored into the calculation.

4.4.3 Maximizing Factor Return. Aiming to enhance investment profitability and model performance, we focus on maximizing factor return. Factor returns, represented as cross-sectional regression coefficients, indicate the return attributable to a specific common factor [10, 26]:

$$\hat{\mathbf{r}}_{t+k} = b_k^t \mathbf{f}_k^t, \quad (17)$$

where $b^t \in \mathbb{R}$ is the factor return at time t . Since a deep factor is expected to have a higher cumulative factor return, the loss for factor return is defined as

$$\mathcal{L}_f = \frac{1}{|\mathcal{K}||\mathcal{T}|} \sum_{k \in \mathcal{K}} \sum_{t \in \mathcal{T}} \psi_k^t d_f^k, \quad (18)$$

where we introduce a local optimizer to learn the parameter ψ_k^t in the later context, which is independent of the global optimizer.

4.4.4 Improving “Attention Estimate”. We use L_2 -norm to evaluate the deviation between the deep factor $\mathbf{f}_k^t$ and its corresponding attention estimate $\hat{\mathbf{f}}_k^t$, which can be calculated as

$$e_k^t = \|\mathbf{f}_k^t - \hat{\mathbf{f}}_k^t\|_2. \quad (19)$$

Therefore, the attention estimate loss is defined as

$$\mathcal{L}_e = \frac{1}{|\mathcal{K}||\mathcal{T}|} \sum_{k \in \mathcal{K}} \sum_{t \in \mathcal{T}} e_k^t. \quad (20)$$

4.5 Optimization

To improve the performance of the E2E framework, besides the global optimizer we further propose a specialized optimizer, named cross-sectional optimizer, for learning the directional cross-section coefficient.

4.5.1 Cross-sectional optimizer. The cross-sectional optimizer γ_1 is a local optimizer for learning the factor return of the deep factor, which is defined as the cross-sectional regression coefficient (factor return) at each time $t \in \mathcal{T}$. A linear regression model is fitted by minimizing the mean squared errors of the predicted returns and the target returns. Since a deep factor has a positive or negative IC, the factor’s return has the same direction as its IC. Therefore, we first determine the direction of the deep factor and then estimate its factor using the cross-sectional optimizer. This cross-sectional optimizer is independent of the global optimizer we use to train our E2E framework.

4.5.2 Global optimizer. The global optimizer γ_2 is designed to update the parameters of our deep E2E framework. It is set to simultaneously optimize the parameters of the factor selection, multifactor model, directed contribution estimator, and portfolio construction module.

5 EXPERIMENT RESULTS

In this section, we evaluate the effectiveness of our proposed E2EAI framework on real-world stock market data. For our evaluation, we select one of the largest emerging markets as our case study. To comprehensively assess the performance of our model, we design a series of experiments aimed at answering the following research questions:

- **Exp 1:** Does our method outperform non-end-to-end approaches in factor investing under diverse portfolio construction settings?
- **Exp 2:** Is the directional recovery component essential for the effectiveness of the E2EAI framework?
- **Exp 3:** How can we interpret the meaning of the deep factor?

5.1 Experiment Settings

Stock Pool. To have a fair comparison to the existing deep multifactor model, we use the same stock pool with the same period used in [26]. The stock pool consist of the most representative stocks of China stock markets, *i.e.*, the constituents of three major China Stock Indices (CSIs): CSI 300, CSI 500 and CSI 1000, which represent more than 80% of total market capitalization of the entire stock market in China [26]. These three indices covers 1800 stocks largest and most liquid stocks in the market at the same time, while they cover more than 2800 stocks in the period from 2010 to mid-2022.

Dataset Construction. The factors consist of fundamental factors and technical factors based on various data sources. The fundamental and market data come from publicly available data from tushare.pro, which can also be obtained from Refinitiv or WIND institutional investor databases [26]. To effectively mitigate survivorship bias, our dataset is constructed via a point-in-time methodology, encompassing even those entities that experienced delisting or total disappearance within the study’s timeframe. The historical

Table 1: The comparison of portfolio performances based on *top-1-of-K* strategy over the period 2015 to mid-2022 (best/2nd best).

Method	CSI 1000					CSI 500					CSI 300				
	$\alpha\uparrow$	IR $\uparrow$	MD $\downarrow$	TT	$\bar{n}$	$\alpha\uparrow$	IR $\uparrow$	MD $\downarrow$	TT	$\bar{n}$	$\alpha\uparrow$	IR $\uparrow$	MD $\downarrow$	TT	$\bar{n}$
Linear	7.79	1.35	0.08	0.76	100	5.62	0.77	0.04	0.89	50	1.73	0.32	0.08	0.89	30
EW	14.65	1.92	0.06	0.71	100	8.91	1.86	<u>0.06</u>	0.97	50	7.57	1.20	<u>0.06</u>	0.71	30
MLP	15.16	<u>3.53</u>	<u>0.07</u>	0.88	100	4.48	0.67	0.05	0.78	50	3.56	0.65	0.15	0.88	30
MGAT	9.24	1.63	0.15	1.05	100	4.96	0.62	0.12	0.83	50	3.17	0.56	0.09	1.53	30
DMFM	<u>16.36</u>	3.46	0.08	0.92	100	9.03	1.89	0.07	1.01	50	7.85	1.09	0.06	0.96	30
E2E _l	15.14	2.92	0.07	0.94	100	<u>12.00</u>	2.18	0.07	1.11	50	<u>9.60</u>	<u>1.46</u>	0.09	0.71	30
E2E _d	18.10	4.35	0.08	0.92	100	13.37	2.45	0.08	1.01	50	13.40	2.02	0.07	0.97	30

Table 2: The comparison of portfolio performances based on auto-construction over the period 2015 to mid-2022 (best/2nd best).

Method	CSI 1000					CSI 500					CSI 300				
	$\alpha\uparrow$	IR $\uparrow$	MD $\downarrow$	TT	$\bar{n}$	$\alpha\uparrow$	IR $\uparrow$	MD $\downarrow$	TT	$\bar{n}$	$\alpha\uparrow$	IR $\uparrow$	MD $\downarrow$	TT	$\bar{n}$
S-Best	7.10	0.42	0.11	0.81	319	0.74	0.90	0.01	0.24	433	-0.06	-0.58	<u>0.01</u>	0.16	272
S-Avg	<u>11.10</u>	<u>0.67</u>	<u>0.11</u>	0.82	321	<u>0.88</u>	<u>1.16</u>	<u>0.01</u>	0.22	455	0.01	-0.19	0.01	0.13	289
S-T20	-0.04	-0.08	0.32	1.75	32	-9.44	-0.81	0.22	1.03	25	11.22	<u>0.69</u>	0.11	0.40	50
E2E _{auto}	20.12	1.37	0.09	0.96	95	8.88	2.47	0.04	0.76	129	<u>7.40</u>	1.56	0.04	0.60	103

Table 3: The comparison of E2EAI over the period 2015 to mid-2022 (best).

Method	CSI 1000					CSI 500					CSI 300				
	$\alpha\uparrow$	IR $\uparrow$	MD $\downarrow$	TT	$\bar{n}$	$\alpha\uparrow$	IR $\uparrow$	MD $\downarrow$	TT	$\bar{n}$	$\alpha\uparrow$	IR $\uparrow$	MD $\downarrow$	TT	$\bar{n}$
E2E _{auto} nd	-1.46	-0.52	0.16	0.81	278	-1.00	-1.50	0.02	0.26	427	0.28	-1.14	0.01	0.18	276
E2E _{auto} ^{nb}	17.47	0.98	0.10	1.04	95	5.27	1.70	0.04	0.59	191	2.94	0.80	0.04	0.63	120
E2E _{auto} ^l	15.14	0.75	0.11	0.93	141	3.71	1.65	0.03	0.70	192	7.24	1.17	0.05	0.49	103
E2E _{auto}	20.12	1.37	0.09	0.96	95	8.88	2.47	0.04	0.76	129	7.40	1.56	0.04	0.60	103

data of these companies were integrated into the database, treating them as active listings up until their ultimate delisting or cessation. This approach ensures our dataset authentically reflects the complexity inherent to the prediction problem. We create our dataset on a daily basis and divide the original factors into seven groups: Value, Growth, Momentum, Quality, Size, Liquidity [14, 16, 26]. The daily stock return is calculated by $r_{t+k} = \frac{p_{t+k} - p_{t+1}}{p_{t+1}}$, where t is the forecast day and p_{t+k} denotes the volume-weighted average price of all stocks on the next k day after the forecast day. To avoid look-ahead bias, we create a point-in-time dataset [26] and delete the intersection of the validation and test sets from the validation set. For the stock graphs, the industry graph is created based on the CITIC Securities Industry Classification Standard [26] and the universal graph is created based on the list of constituent members issued by the China Securities Index Company. In creating the dataset, we use the time-series cross-validation technique, which divides the entire dataset into 14 groups in chronological order [26].

Baselines and Model Implementation. We compare our proposed method with the following baselines: 1) **Linear**[6, 18] is a linear multifactor model. 2) **EW** [14] is an equally weighted model that has nothing to do with data mining or in-sample optimization. In

practice, any deviations from equal weighting are determined by economic rationality [14]. 3) **MLP** is a nonlinear model that learns factors from the stock context based on neural networks and consists of a context encoder and a feature decoder. 4) **MGAT** [26] is a model based on a context encoder, a GAT, which learns the interactions between stocks in a universe graph, and a decoder, which learns a deep factor from the stock context and the universal relationships of the stocks. 5) **DMFM** [26] is a deep multifactor model. For comparison of E2E and step-by-step methods, we provide three additional step-by-step baselines with different criteria for factor selection: 1) **S-Best**, which selects the factors with the best performance at different prediction horizons; 2) **S-Avg**, which selects the factors that can perform well on average at different prediction horizons. 3) **S-T20** selects only the factors that perform well in predicting one-month returns.

5.2 Exp1: Effectiveness of E2EAI

In this experiment, we evaluate the effectiveness of our E2EAI framework within the scope of factor investing. We aim to answer two crucial questions concerning its effectiveness and performance:

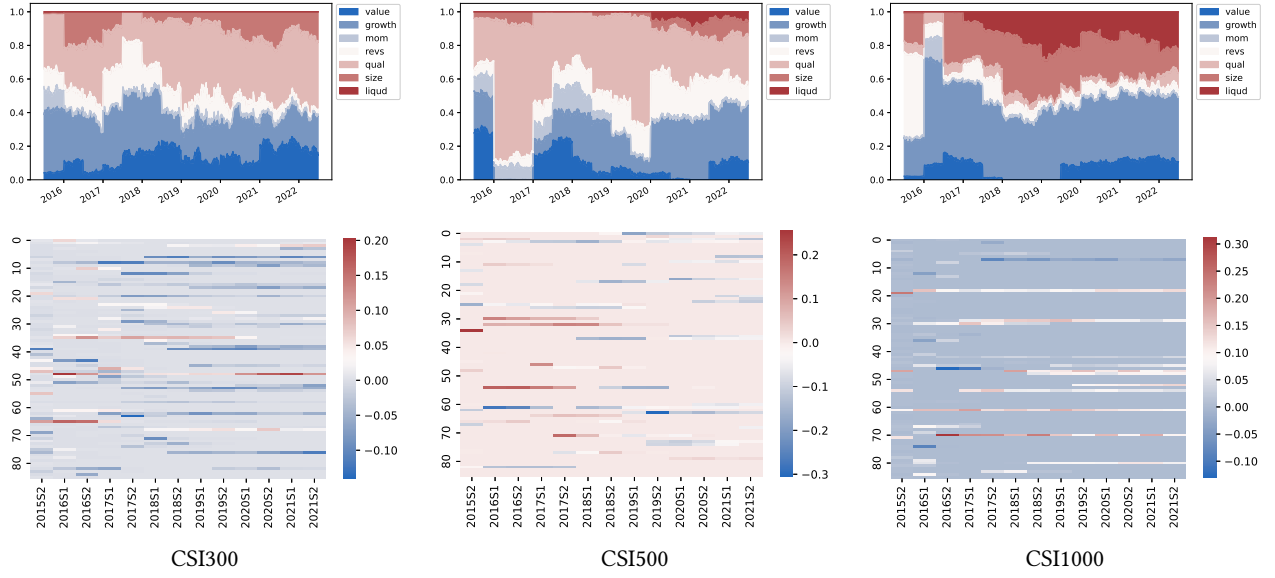


Figure 2: The heat map shows the dynamic contribution of the original factors to the deep factors: 1) the first row shows the dynamic attention allocation to the original factor groups; 2) the second row shows the average attention weights for each style factor on a semi-annual basis.

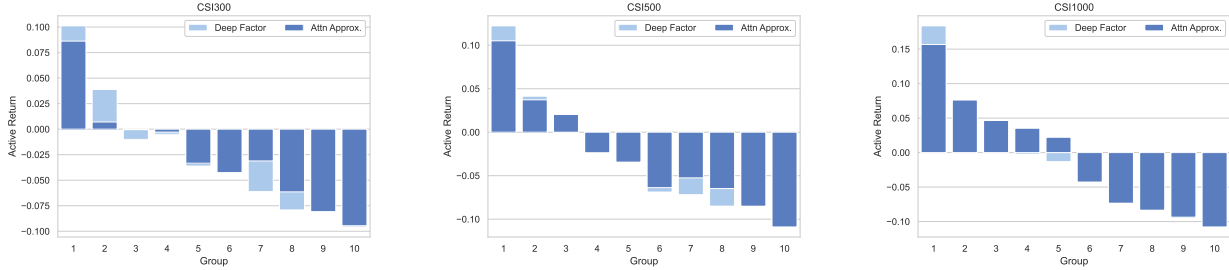


Figure 3: The monotonicity analysis of the deep factor and its attention approximation (AA) on different stock universes. 1) The average returns increase monotonically in stratified groups based on factor exposure, showing good stability of their predictive power; 2) the AA of deep factor can linearly explain most cases in CSI300 and CSI500 from the perspective of portfolio performance in different groups.

- **Exp 1a:** Does the deep factor, derived from our end-to-end learning approach, exhibit superior effectiveness compared to those obtained from non-end-to-end methods, given the same portfolio construction strategy?
- **Exp 1b:** Does our E2EAI surpass traditional automated portfolio construction pipelines in terms of effectiveness?

We use three widely used metrics for active investing: excess return (α), information ratio (IR), and maximum active drawdown (MD), as well as two additional descriptive metrics such as buy and sell turnover (TT) and average number of stocks held. $\uparrow$ and $\downarrow$ denote the metrics “the higher the better” and “the lower the better” respectively.

5.2.1 End-to-End Learning Empowers Deep Factor. To address **Exp 1a**, we examine the stock selection ability of the deep factor learned via end-to-end learning, comparing it with other non-end-to-end methods. The subsequent portfolio construction strategy we employ

is the *Top-1-of-K Strategy*. This popular method is used to test a factor’s ability to select stocks and to demonstrate the effectiveness of a candidate model. It involves dividing stocks into K groups *w.r.t.* factor scores, then constructing an equally-weighted portfolio based on the group with the largest factor scores (assuming the factor is positively linearly correlated with expected return and vice versa). Table 1 shows the comparison of different methods based on a stratification of 10 groups. Here, $E2E_d$ and $E2E_l$ represent the portfolio construction based on a deep factor and its linear approximation (AA) via attention estimate, respectively.

5.2.2 End-to-End Outperforms Step-by-Step. In Table 2, we compare our E2EAI framework with stepwise baselines. These baselines employ different factor selection paradigms but utilize the same portfolio construction module as described in Section 4.3. The results show that our E2EAI outperforms the stepwise baselines, thus

highlighting the suboptimal issue caused by stepwise automatic construction. Overall, our E2EAI demonstrates superior performance over different baselines under various active investing paradigms, with respect to metrics such as excess return and information ratio.

5.3 Exp2: Directional Recovery is Necessary

This experiment aims to determine whether directional recovery is necessary in the E2EAI framework. Directional recovery was initially proposed to address directional conflicts during multitask learning. We introduce several variants in this experiment:

- To test the necessity of directional recovery, $E2E_{auto}^{nd}$ is an E2EAI variant without directional recovery (directional buffer) of the deep factor (see Line 7-8, Algorithm 1).
- To examine the influence of keeping a bias term in estimating factor returns, $E2E_{auto}^{nb}$ is an E2EAI variant without a bias term in the cross-sectional model for learning the cross-sectional coefficient (see Line 12, Algorithm 1).
- To assess the effectiveness of the linear approximation of the deep factor in active investing, $E2E_{auto}^l$ is an E2EAI variant where automatic portfolio construction is based on the linear combination of the attention approximation.
- $E2E_{auto}$ represents the comprehensive E2EAI framework proposed in this work.

The ablation study of these E2EAI variants is presented in Table 3. Without directional recovery, $E2E_{auto}^{nd}$ performs worse than the other end-to-end variants as shown in Table 3, underscoring the necessity of directional recovery. Similarly, the performance drop in $E2E_{auto}^{nb}$ compared to $E2E_{auto}$ signifies the importance of keeping a bias term in estimating factor returns. Lastly, the comparable performance of $E2E_{auto}^l$ and $E2E_{auto}$ in the CSI 300 context suggests the effectiveness of the linear approximation approach.

5.4 Exp3: Interpreting Deep Factors

5.4.1 Directional Contribution Reveals Dynamic Allocation of Original Factors. Fig. 2 presents the group contribution and the directional contribution of each factor over the test period. A notable observation relates to liquidity: it contributes minimally in the CSI300 universe, while in CSI500, it plays a significant role in the second tier of the most liquid stocks with the largest market capitalization. However, liquidity's contribution increases in CSI1000, which comprises 1000 small-cap companies that are less liquid than those in CSI800 (CSI300 and CSI500). This indicates that liquidity is more critical in small-cap stocks, aligning with practical observations.

5.4.2 Monotonicity Analysis of Deep Factor and its Attention Approximation on Different Stock Universes. The portfolio $E2E_l$, constructed by attention approximation over attention estimate, displays a monotonic similarity to $E2E_d$, as shown in Fig. 3. This reveals the robustness of their predictive power, where the attention approximation of deep factor can linearly explain most cases in CSI300 and CSI500 from the perspective of portfolio performance in different groups.

5.4.3 E2E Linear Approximation Displays Comparable Performance in Large-Cap Stocks. Referring back to the automatic construction case discussed in Section 5.3, we observe a similar pattern here. The comparable performance of $E2E_{auto}^l$ and $E2E_{auto}$ within the

CSI 300 context, which represents the 300 largest and most liquid on-listed stocks in the market, suggests a noteworthy observation. Although the stock selection ability of the deep factor $E2E_d$ is superior to that of the linear factor $E2E_l$ (as shown in Table 1), their similar automatic construction performance under the end-to-end framework indicates that a deep portfolio module (see Eq. 11-12) may introduce nonlinear effects to $E2E_{auto}^l$'s portfolio construction to some extent.

6 CONCLUSION

In this paper, we pioneer the introduction of an end-to-end (E2E) framework for active investing, demonstrating superior performance over existing strategies across three major stock universes in the world's largest emerging stock market. Our novel incorporation of a directional buffer plays a critical role in alleviating directional conflicts inherent in E2E multi-task learning, paving the way for successful E2E active investing. Furthermore, our approach enables the discernment of both positive and negative contributions from each factor, facilitating the dynamic allocation of each original factor. In our future work, we aim to broaden the application of our methodology to global markets, focusing on diverse investment objectives and sector-specific characteristics.

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